Yufeng Chen

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Education

Ph.D. Candidate, Computer Science — Sep. 2021 – Present
Purdue University, West Lafayette, USA — GPA: 3.83/4.0

Bachelor of Engineering, Electronic Information Engineering — Sep. 2017 – Jul. 2021
University of Science and Technology of China, Hefei, China — GPA: 3.87/4.3
Talent Program in Information Science and Technology

Research & Engineering Projects

Efficient Streaming of Immersive Content over Wi-Fi Networks — 2022 – Present
Ph.D. Thesis Research, Purdue University

- *Progressive loading for networked VR:* Designed packet-level progressive streaming for static and dynamic point clouds; evaluated multi-stream QUIC and SCTP out-of-order delivery to mitigate head-of-line blocking and improve rendering smoothness.
- *Scalable multi-user VR:* Built a commodity-Wi-Fi multicast system to keep transmission overhead independent of group size; validated with 23 users in groups of up to 16.
- *Adaptive redundancy scheduling for multi-user VR:* Designed layered 3D Gaussian Splatting packetization and a belief-driven scheduler for fair, robust multi-user streaming over lossy Wi-Fi.
- *Real-time VR traffic scheduling:* Formulated an end-to-end, slack-aware scheduling framework that combines receiver-specific latency budgets derived from application semantics with observed uplink delays; evaluated through ns-3 simulations and an OpenWrt testbed.

QoE-based Routing Path Selection — 2022 – 2023
Purdue University and Juniper Networks

- Analyzed multi-scale burstiness in 12 video services and 7 backbone traces; designed a routing path selection policy based on application-level (QoE) metrics.
- Evaluated DASH and WebRTC workloads in a Mininet/CloudLab testbed; improved streaming QoE by 14% on average and reduced conferencing packet latency by 11% versus baseline path-selection approaches.

Scheduling Diverse QUIC Streams — 2022 – 2023
Purdue University

- Extended Cloudflare's `quiche` with FIFO, static round-robin, DTP-based dynamic-priority, and weighted round-robin schedulers across reliable QUIC streams and unreliable datagrams.
- Built CloudLab benchmarks measuring CPU, memory, and sender-side queueing delay; characterized the scalability costs of stream creation and dynamic-priority scheduling tradeoffs.

Multi-Domain Optical Network Testbed with DeepRMSA — Jun. 2020 – Nov. 2020
University of California, Davis

- Developed a deep reinforcement learning-based routing, modulation, and spectrum assignment system for hierarchical multi-domain elastic optical networks.
- Integrated Actor-Critic agents with ONOS SDN controllers; processed 1.97M requests in 80.9 hours, reducing broker blocking from 23% to 3% while maintaining 148.87 ms average latency.

RDMA Optimization for MongoDB — Jun. 2018 – Nov. 2018
University of Science and Technology of China

- Replaced TCP/IP-based MongoDB cluster data transfer with RDMA to reduce client–server and inter-server communication latency.
- Achieved up to 3.58×/6.16× peak Put/Get throughput improvement over the TCP/IP baseline.

Publications

- Yuqi Zhou, Shuqi Liao, **Yufeng Chen**, Sonia Fahmy, and Voicu Popescu. "Scalable Collocated Multi-User VR Through Virtual Environment User Spatial Coherence." *IEEE Transactions on Visualization and Computer Graphics*, 2026. Also in Proc. IEEE VR, Mar. 2026.
- **Yufeng Chen**, Umakant Kulkarni, Voicu Popescu, and Sonia Fahmy. "RUN: A Case for Cross-Layer Networked Virtual Reality." *Proceedings of the 33rd ACM International Conference on Multimedia*, MM '25, pp. 12111–12120, Oct. 2025.
- Umakant Kulkarni, **Yufeng Chen**, Patrick Melampy, and Sonia Fahmy. "Toward QoE-based Routing Path Selection." *IEEE HPSR*, pp. 114–119, Jun. 2023.
- **Yufeng Chen**, Akhil Prasad, Sonia Fahmy, and Voicu Popescu. "Scheduling Diverse QUIC Streams." Poster, USENIX NSDI, Apr. 2023.

Teaching Experience

- Teaching Assistant, CS 536: Data Communication and Computer Networks — 2024, 2025
- Teaching Assistant, CS 240: Programming in C — Spring 2022
- Teaching Assistant, CS 251: Data Structures — Fall 2021

Technical Skills

Programming: Python, C/C++, Java, Shell

Networking & Systems: QUIC, TCP/IP, Wi-Fi, multicast/broadcast, socket programming, Linux, RDMA

VR & Graphics: Unity, point clouds, 3D Gaussian Splatting

Tools: Git, Docker, Wireshark, ns-3, ONOS